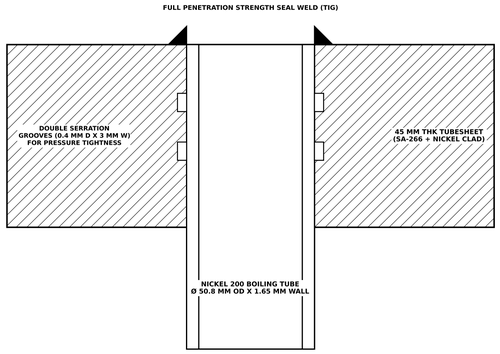
TRIPLE-EFFECT NAOH EVAPORATOR — MECHANICAL FABRICATION & PRESSURE VESSEL DETAILS

CLOSE-UP ENGINEERING DETAILS: TUBE JOINTS, DISHED HEAD FORMING, SKIRT BASE CHAIR, DEMISTER & TEMA PARTITIONS

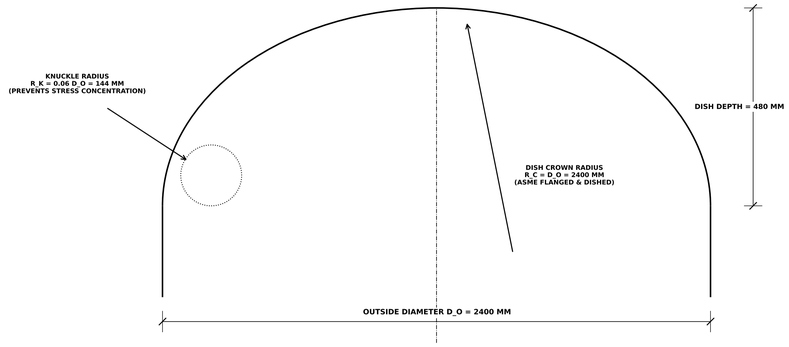
DETAIL A — TUBE-TO-TUBESHEET JOINT (ASME SEC VIII DIV 1 UW-20)

ROLLER EXPANDED WITH DOUBLE SERRATION GROOVES & STRENGTH SEAL FILLET WELD



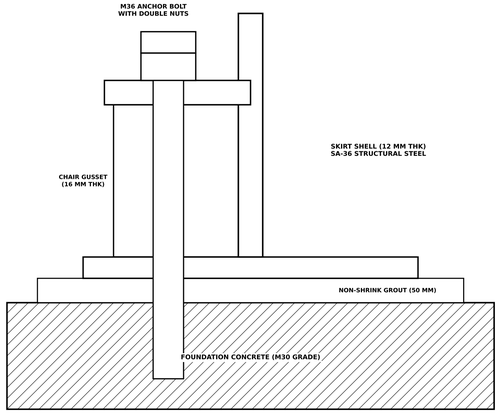
DETAIL B — ASME TORISPHERICAL DISHED HEAD FORMING GEOMETRY

STANDARDIZED ASME CODE RATIOS (CROWN RADIUS R_C = D_O, KNUCKLE RADIUS R_K = 0.06 D_O)



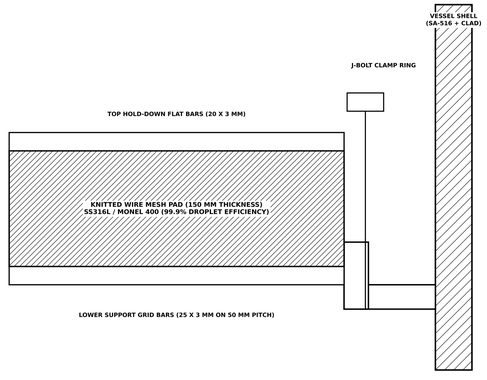
DETAIL C — VESSEL SKIRT SUPPORT & ANCHOR BOLT CHAIR DETAIL

FABRICATED STEEL ANCHOR BOLT CHAIR ON CONCRETE FOUNDATION SLAB



DETAIL D — DEMISTER PAD CLAMPING & SUPPORT GRID DETAIL

KNITTED SS316L/MONEL WIRE MESH WITH LOWER GRID & TOP HOLD-DOWN BARS



OPTIONAL DIAGRAM 3 — EQUIPMENT MECHANICAL FABRICATION & ASME SECTION VIII DETAILS

DRAWING TITLE MECHANICAL FABRICATION & JOINT DETAILS	SUBJECT PROCESS HEAT TRANSFER (PHT)	COURSE / BRANCH B.E. CHEMICAL ENGINEERING	DESIGN CODES ASME SEC VIII DIV 1 / TEMA CLASS R
STUDENT NAME SOURISH SENAPATI	EXAM ROLL NUMBER 002310301035	ASSIGNED ROLL (LAST 2 DIGITS) 35 (TARGET: 50.0 WT% NAOH PRODUCT)	DATE 01 / 09 / 2026
PRESSURE VESSEL CODE ASME SECTION VIII DIV 1 (UW-20 / UG-32)	WELD JOINT EFFICIENCY E = 1.0 (100% FULL RADIOGRAPHY)	CORROSION ALLOWANCE / CLAD 3.0 MM CAUSTIC CORROSION ALLOWANCE + NICKEL CLAD	MATERIALS OF CONSTRUCTION NICKEL 200, SS316L, SA-516 GR.70, SA-266 CL.2

ALL WELDING PROCEDURES: ASME SECTION IX (WPS/PQR) | TUBE-TO-TUBESHEET: ROLLER EXPANDED + STRENGTH SEAL WELD | DIMENSIONS: MM